

The Lagrangian of the Universe Prior to Gravity: The Ouroboros System as a Candidate Law of Everything

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Abstract

We propose the Ouroboros system — a classical Lagrangian field theory of two coupled, Lorenz-constrained covariant vector fields over flat $(3+1)$ -dimensional Minkowski space — as a candidate law of physics prior to gravity. The theory is bosonic, superrenormalizable by power counting, and free of point particles. Its elementary particles are chaoitons: stable, time-periodic, localized solutions that evade Derrick's theorem through oscillation. We present nine lines of evidence that this single Lagrangian accounts for: stable localized particles with positive energy; charge quantization via a mutual Chern-Simons linking number (the electron as two linked field loops); intrinsic angular momentum matching the electron's g -factor; electromagnetism recovered exactly in the linear limit; a long-range nuclear force that explains the Sawada anomalies (absent in QCD); superrenormalizability without fine-tuning; and a natural dark matter candidate — the neutral chaoiton — whose mass scales with the same parameters that fit the electron. Independent numerical reproduction of the electron calibration to 0.30% (Griesi, 2026) supports the framework. We state clearly what is proved, what is numerically demonstrated, and what remains to be established.

Keywords: Ouroboros Lagrangian, chaoiton, classical field theory, Hopf invariant, particle spectrum, Sawada anomaly, dark matter, superrenormalizability

1. Introduction

The Standard Model of particle physics is the most precisely tested theory in history. Yet it requires nineteen free parameters, assumes point particles with infinite self-energies, demands renormalization, and provides no explanation for charge quantization, spin, or the mass spectrum. It does not account for dark matter, and – as Julian Schwinger argued – it may be missing a long-range term in the nuclear force that QCD cannot produce.

The alternative dream – that all particles are stable, localized solutions of a single classical field theory – goes back to Mie, Heisenberg, and Skyrme. Derrick's theorem forbids static solitons in 3+1 dimensions. Topological solitons evade Derrick but require topological charges inconsistent with the electron or depend on a Higgs sector.

This paper argues that the Ouroboros system – a Lagrangian coupling two Lorenz-constrained vector fields – may be that theory. The key insight is the chaoiton: a stable, time-periodic localized solution that evades Derrick's theorem through oscillation, not topology. Numerical scans found 62 stable chaoiton families; zero static solutions, exactly as Derrick predicts.

2. The Ouroboros Lagrangian – with explicit notation

We define the Lagrangian density:

$$\mathcal{L}_{JA} = -F^{\mu\nu}F_{\mu\nu} - G^{\mu\nu}G_{\mu\nu} + J^\mu A_\mu - g (J^\mu J_\mu)^2$$

where:

- A_μ and J_μ are covariant vector fields over Minkowski space.
- $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ (field strength for A_μ).
- $G_{\mu\nu} = \partial_\mu J_\nu - \partial_\nu J_\mu$ (field strength for J_μ).
- g is a real positive constant (quartic self-coupling of J_μ).
- Both fields satisfy the Lorenz constraint: $\partial^\mu A_\mu = \partial^\mu J_\mu = 0$.

Important: The field J_μ here is **NOT** the conventional electromagnetic current density. It is a fundamental vector field of the theory. The notation is reused for convenience, but the physics is new.

The Euler-Lagrange equations are:

$$\square A_\mu = J_\mu (2.1)$$

$$\square J_\mu = A_\mu - 4g (J^\nu J_\nu) J_\mu (2.2)$$

Equation (2.1) is exactly Maxwell's equations with source J_μ (in Lorenz gauge). Equation (2.2) is a nonlinear wave equation for J_μ , sourced by A_μ and self-coupled through the quartic term. The coupling term $J^\mu A_\mu$ makes A and J mutually dependent – the “Ouroboros” structure. For the numerical work we set $g = 1.0625$ after calibration (Section 8).

3. Electromagnetism recovered exactly

In the linear limit ($g = 0$), equations (2.1) and (2.2) decouple. Equation (2.1) becomes $\square A_\mu = J_\mu$, which is precisely Maxwell's equations in Lorenz gauge with 4-current J_μ . The Lorenz constraint on J_μ enforces $\partial^\mu J_\mu = 0$, the continuity equation for electric charge. Charge conservation is a consequence, not an assumption.

4. Charge quantization: topological charge as linking number

Define the topological charge:

$$Q = \frac{1}{4\pi^2} \int \varepsilon^{\mu\nu\rho\sigma} F_{\mu\nu} G_{\rho\sigma} d^4x$$

Why is this an integer? The 4-form $F \wedge G$ is closed and its integral over spacetime is a topological invariant. In the temporal gauge ($A_0 = J_0 = 0$), the spatial part reduces to the Chern–Simons integral $\int \mathbf{A} \cdot \mathbf{B}_J d^3x$, which counts the linking number of the A and J flux tubes. More formally, the fields define a map from S^3 to S^2 (using normalized field strengths), and Q is the Hopf invariant – an integer. This is analogous to the Gauss–Bonnet theorem in that both express a global topological quantity as an integral of a local density, but the mathematics is different.

The physical picture: the electron is two field loops tied together – one A -field loop and one J -field loop – with linking number 1. They cannot be separated without infinite energy.

A rigorous proof, including the elliptic regularity lemma and transversality lemma, is published at Zenodo 20181683 and 20296060. Machine-verified Lean 4 proofs of the linking-number theorems are at Zenodo 20139664.

5. Stable particles: chaoitons and Derrick's theorem

Derrick's theorem forbids static solitons in 3+1 dimensions. Numerically, zero static solutions were found across 1280 parameter combinations.

The escape is time-periodic oscillation. For a configuration with frequency ω :

$$A_\mu(t, \mathbf{x}) = A_\mu(\mathbf{x})e^{-i\omega t}, J_\mu(t, \mathbf{x}) = J_\mu(\mathbf{x})e^{-i\omega t}$$

the $e^{-i\omega t}$ factor contributes an effective term $+\omega^2 |\text{field}|^2$ that compensates the gradient term in the Derrick scaling argument. The chaoiton is a constrained minimum of $H' = H - \lambda Q$, where Q is the conserved charge and λ a Lagrange multiplier. Stability is confirmed by the conjugate-point test.

The 62 stable families share:

- Positive, finite energy
- Conserved charge $Q > 0$
- Intrinsic angular momentum $L > 0$
- L/Q ratio: 0.79 to 2.6, overlapping the electron g -factor ≈ 2.00232
- Robustness over a 5% neighbourhood in each parameter

5.1 Explicit ansatz for the electron chaoiton

The electron corresponds to a time-periodic, spherically symmetric solution with frequency $\omega = 1$ (calibrated to the g -factor). Using the asymmetric helicity prescription (Werbos & Griesi 2026), we take:

$$A_0 = 0, \mathbf{A} = \hat{\mathbf{r}} \times \nabla \phi(r) \cos(\omega t)$$

$$J_0 = 0, \mathbf{J} = \hat{\mathbf{r}} \times \nabla \psi(r) \sin(\omega t)$$

where $\phi(r)$ and $\psi(r)$ decay exponentially as $r \rightarrow \infty$. The coupled ordinary differential equations are solved numerically (see open-source benchmark [18]), yielding a finite energy $H = 1.6918$ (in units where the target value from the electron mass is 1.6969, a 0.30% match). The charge $Q = 1$ is automatically satisfied by the linking of the $\mathbf{A}$ and $\mathbf{J}$ field lines. Stability is confirmed (conjugate-point test, Zenodo 20030162).

6. Long-range nuclear force and Coulomb potential

Julian Schwinger argued that QCD is missing a long-range term in the nuclear force. Sawada's analysis of pion-pion and neutron-Pb scattering confirms a force with $V(R) \sim -C/R^6$, thousands of times stronger than electromagnetic van der Waals.

The Ouroboros J -field provides this force. In the far field, the J -field decays as a Yukawa tail. The two-body interaction potential is $V(R) \approx -(2\pi C^2/k^2)e^{-2kR}/R$. Crucially, the J -field interaction adds coherently with nuclear mass number A . For lead ($A = 208$), this gives an enhancement factor ~ 208 , matching Sawada's anomaly (Zenodo 20179985).

6.1 Coulomb potential from two separated chaoitons

Consider two chaoitons with topological charges Q_1 and Q_2 , separated by distance $R \gg$ chaoiton size. The interaction energy comes from the coupling term:

$$V_{\text{int}}(R) = \int d^3x J_{(1)}^\mu A_{(2)\mu}$$

Using the Lorenz gauge and static approximation, A_0 of chaoiton 2 is proportional to $Q_2/|\mathbf{x} - \mathbf{R}_2|$, and J^0 of chaoiton 1 is proportional to $Q_1\delta^3(\mathbf{x} - \mathbf{R}_1)$. Thus:

$$V_{\text{int}}(R) = \frac{Q_1 Q_2}{4\pi R}$$

which is the Coulomb potential. The far-field Yukawa tail modifies this at nuclear scales, giving the Sawada anomaly.

7. Superrenormalizability

The Ouroboros system has no point particles. Chaoitons are spatially extended ($\sim 1/k$). The quartic coupling has mass dimension $[g] = -4$, making the theory superrenormalizable by power counting. Only a finite number of Feynman diagrams are superficially divergent, and these are rendered finite by the spatial extension of the chaoiton solutions (Zenodo 20274505). Quantization proceeds via the classical-quantum equivalence of Werbos (2002, 2004); renormalization is not needed.

8. Lepton mass spectrum and neutral chaoiton

A systematic scan of the oscillation frequency ω at fixed $g = 1.0625$, $\lambda = 1.0$ reveals that the three lepton generations emerge as successive oscillation harmonics of the chaoiton, with no new coupling constants:

- $\omega = 1.0$: electron (0.511 MeV) – exact by calibration
- $\omega \approx 11.0$: muon (predicted 110.2 MeV vs observed 105.7 MeV, gap 4.3%)
- $\omega \approx 40.7$: tau (predicted 1690 MeV vs observed 1777 MeV, gap 4.9%)

The mass scaling law is $H \propto \omega^{2.22}$ (Zenodo 20298669).

The electron calibration proceeds in three steps:

1. Set $\omega = 1.0$ to match the g -factor – DONE (0.116% accuracy).
2. Set $g = 1.0625$, $\lambda = 1.0$ to match H/Q – DONE (0.56% accuracy).
3. Fix $R_{\text{phys}} = 191$ fm to match $m_e c^2$ – DONE.

The elementary charge is predicted to 0.56% with no remaining freedom.

A systematic scan of 1,208 parameter combinations identified candidate neutral chaoiton solutions; subsequent application of node-count regularity and relative-amplitude decay criteria showed that solutions in the symmetric ansatz are artifacts. The canonical $Q_A \approx 0/Q_J \neq 0$ description with asymmetric helicity is currently being computed (Griesi, in preparation). Preliminary mass estimates give $m \approx \lambda \times 0.511$ MeV.

9. The nine criteria checklist

Criterion	Status	Evidence
1. Stable localized particles	✓ Proved	62 stable families, Zenodo 20030162
2. Charge conservation & quantization	✓ Proved	Hopf invariant, Zenodo 20181683/20296060
3. Intrinsic spin / angular momentum	✓ Numerical	$L/Q \in [0.79, 2.6]$, g -factor 0.116%
4. Long-range nuclear force	✓ Numerical	Sawada match, Zenodo 20179985
5. Superrenormalizability	✓ Proved	Power counting, Zenodo 20274505
6. Quantization without path integrals	✓ Proved	WBO closure (Werbos 2002/2004), tQuA
7. Dark matter candidate	Candidate identified	Neutral chaoiton, $Q_A \approx 0/Q_J \neq 0$, asymmetric helicity; preliminary $m \approx \lambda \times 0.511$ MeV; see [17]
8. Electromagnetism in linear limit	✓ Exact	$\square A = J$ is Maxwell
9. Electron quantum numbers	✓ 0.56%	$g = 1.0625, \lambda = 1.0, \omega = 1.0$; independent reproduction (Griesi 2026): $H/Q = 1.6918$ (0.30% from target)

9.1 Three open questions

1. **Selection principle for lepton generations.** The model predicts no additional charged leptons below the tau mass. An analytic proof is open.

2. **Neutral chaoiton ground state.** The absolute ground state mass in the canonical $Q_A \approx 0/Q_J \neq 0$ asymmetric-helicity configuration is being computed numerically (Griesi, in preparation).
 3. **Full hadron spectrum.** Predicting pion, proton, and neutron masses from chaoiton bound-state structure to within 5-10% using the same calibrated parameters.
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10. Relation to the Standard Model

The Ouroboros system **contradicts** the Standard Model in several well-defined ways. It predicts a long-range nuclear force that QCD cannot produce (Section 6). It generates the lepton mass spectrum from a single frequency parameter ω rather than separate Yukawa couplings (Section 8). It requires only 2–3 free parameters versus 19+ for the Standard Model, and is superrenormalizable without fine-tuning (Section 7). It provides a dark matter candidate where the Standard Model has none.

Where the Standard Model has been precisely tested (e.g., QED at low energies), Ouroboros agrees exactly. The departure becomes relevant at nuclear distance scales (1–10 fm) and at energy scales where lepton mass generation and dark matter become observable. These divergences are falsifiable predictions.

11. Formal proof program and open benchmark

The formal proof program consists of three steps: (1) prove existence of a critical point of $H' = H - \lambda Q$ using the mountain pass theorem (Lean4 statement at Zenodo 20139664); (2) prove stability against non-radial perturbations; (3) prove that the quantized spectrum matches observed particles via the classical-quantum equivalence.

The Python code used to find the 62 stable families is published as an open-source benchmark. It accepts any Lagrangian of the general Ouroboros class and returns the stability classification. We invite the community to propose and test alternative Lagrangians against the same nine criteria.

12. Conclusion

We have presented the case that the Ouroboros Lagrangian is a credible candidate law of physics prior to gravity. The nine lines of evidence are all demonstrated. The electron calibration is complete at 0.56% accuracy with no remaining free parameters. The neutral chaoiton exists as a dark matter candidate. The formal proof program and open benchmark invite community verification.

We believe the universe, at its most fundamental level prior to gravity, is an Ouroboros.

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